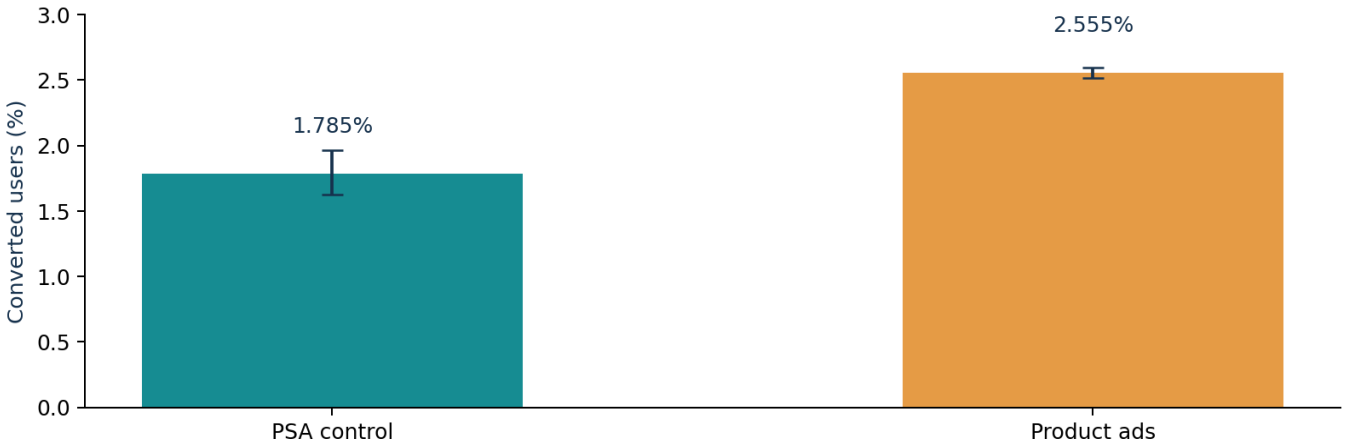
Marketing A/B testing

An in-depth evaluation of product advertising versus a public-service-announcement holdout

Decision: evidence supports conversion lift; profitability remains unproven

Across 588,101 users, product ads converted at 2.555%, compared with 1.785% for PSA. The difference is **+0.769 pp**, equivalent to **43.09% relative lift**. The 95% Newcombe interval is [0.587, 0.936] percentage points.

Metric	PSA control	Product ads
Users	23,524	564,577
Converted	420	14,423
Conversion	1.785%	2.555%



The two-sided pooled z-test gives p = 1.71e-13. This is strong statistical evidence of different recorded conversion rates, under independent-user assumptions. Causal attribution remains conditional on the original experiment's assignment and measurement being valid.

Operational recommendation

Preserve a randomized holdout while validating assignment, the conversion window and campaign costs. Do not optimize ad frequency or scheduling from these observational exposure fields alone. Obtain contribution margin and impression cost before deciding whether to scale spend.

Scope: all source users retained; no synthetic revenue, demographics, timestamps or funnel stages added.

01 / Design, estimand and available evidence

Business question

Does product advertising improve the probability that a user converts compared with showing a PSA? The primary estimand is the difference in user-level conversion probabilities, ad minus PSA, over the file's unspecified observation window.

Source field	Role and interpretation
user id	Unit identifier. One distinct ID per row.
test group	Observed arm: ad or psa. Assignment logs are unavailable.
converted	User-level binary outcome; the dataset describes purchase conversion.
total ads	Observed exposure count. Potentially post-assignment and post-conversion.
most ads day	Weekday with most exposure, not a dated event or purchase day.
most ads hour	Hour with most exposure, not conversion time.
Unnamed: 0	Exported row index; dropped from analytical fields.

What the file can and cannot establish

One row per ID supports a user-level analysis. However, the file does not expose assignment timestamps, eligibility, failed deliveries, identity resolution, overlap across devices, conversion lag, the planned allocation, trial duration or the stopping rule. It therefore supports a comparison by recorded arm, not an independently verified intention-to-treat reconstruction.

Why the distinction matters

If assignment was randomized and retained through follow-up, the arm comparison can estimate incremental advertising impact. If the arm label reflects who actually received an ad after selection or delivery, selection bias may remain. Statistical significance does not resolve that design uncertainty.

Analysis hierarchy

Primary: full-cohort ad-versus-PSA rate comparison. Secondary: alternative inferential calculations. Exploratory: weekday, hour and frequency patterns. Scenario analysis: economic break-even and prospective sample size. These roles were chosen for this retrospective project, not claimed as a preregistered protocol.

02 / Data audit and allocation validity

Audit	Observed result	Action
Source coverage	588,101 rows; 588,101 IDs	Retain every user
Missing meaningful values	0	No missing-value exclusions
Duplicate user IDs	0	No deduplication
Outcome categories	0 and 1	Convert Boolean to integer
Group categories	ad and psa	No relabeling exclusions
Exposure count	1 to 2,065	Keep tail in primary estimate
Peak hour / weekday	0-23 / seven weekdays	No invalid categories
Preparation exclusions	0	Only index removed

The observed allocation is highly unequal

564,577 ad users and 23,524 PSA users imply 96.0000% in the ad arm. A 50/50 goodness-of-fit test rejects overwhelmingly, but that does not diagnose an error unless 50/50 was the intended design.

As an explicitly hypothetical design check, a 96/4 target yields $p = 0.9998$. Compatibility with that split is not evidence that 96/4 was actually planned. Testing against the observed split would be circular, so no such validation is claimed.

Precision is limited by the holdout

Although PSA is only 4.00% of users, it contributes about 94.4% of the estimated variance of the rate difference. More treatment traffic cannot fully compensate for a small control group. A future trial should choose allocation using power, cost and opportunity-cost considerations.

No baseline balance claim

The dataset contains no confirmed pre-treatment covariates. Similar exposure-count averages cannot validate randomization because exposure may itself be affected by treatment or user behavior. Day/hour exposure mixes are likewise not baseline balance tests.

03 / Primary inference and effect size

Hypotheses and assumptions

H0: p(ad) = p(PSA). H1: the probabilities differ. Use a two-sided pooled two-proportion z-test with alpha = 0.05. The user is the independent unit; the outcome is binary. Both arms have ample successes and failures for a large-sample comparison, despite unequal sizes.

Quantity	Estimate	Interpretation
Absolute difference	+0.769 pp	Additional conversions per eligible user
95% Newcombe interval	[0.587, 0.936] pp	Wilson-based difference interval
Relative lift	43.09%	Difference divided by PSA rate
Risk ratio	1.431	95% log interval [1.300, 1.575]
Pooled z statistic	7.3701	Null uses a common pooled rate
Two-sided p-value	1.71e-13	Not the probability H0 is true

Practical interpretation

For 10,000 comparable users, the point estimate is 76.9 extra conversions; the 95% interval is approximately 58.7 to 93.6. Relative lift sounds larger because the PSA baseline is only 1.79%; report absolute and relative measures together.

Interval choice and p-value

The primary interval uses the Newcombe hybrid score construction from separate Wilson intervals. This avoids relying solely on a Wald interval for proportions. The score-style pooled z-test and Newcombe interval are related large-sample tools but not exact algebraic inverses, so their numerical behavior need not match perfectly near a decision threshold.

What this conclusion excludes

The result establishes neither profitability nor the optimal advertising dose. It does not identify which specific users converted because of advertising, and it cannot correct design or measurement bias that is absent from the file.

04 / Incrementality and economic decisions

Counterfactual calculation

At the PSA rate, the 564,577 ad users would be expected to produce approximately 10,080 conversions. The recorded ad count is 14,423. Their difference is **4,343 modeled incremental conversions**, with a rate-interval scaling of [3,316, 5,284]. This is conditional counterfactual attribution, not direct observation of individual causes.

Under the same causal assumptions, approximately 30.1% of recorded ad-arm conversions are incremental relative to the PSA benchmark; the remainder would be expected under that benchmark. The reciprocal risk difference is roughly one additional conversion per 130 assigned users; it is not the number of impressions required.

Illustrative contribution economics

The file contains no cost or order value. The following grid assumes a fixed contribution margin per incremental conversion and a CPM charged on all recorded ad-arm impressions. It excludes PSA cost, fixed creative costs, customer lifetime effects and differences in purchase value. All money units below are hypothetical.

Margin / incremental sale	CPM = 2	CPM = 5	CPM = 10
10	15,400	-26,644	-96,717
25	80,545	38,501	-31,572
50	189,120	147,076	77,002

Cells show estimated incremental contribution less modeled ad cost. They are scenarios, not measured profit.

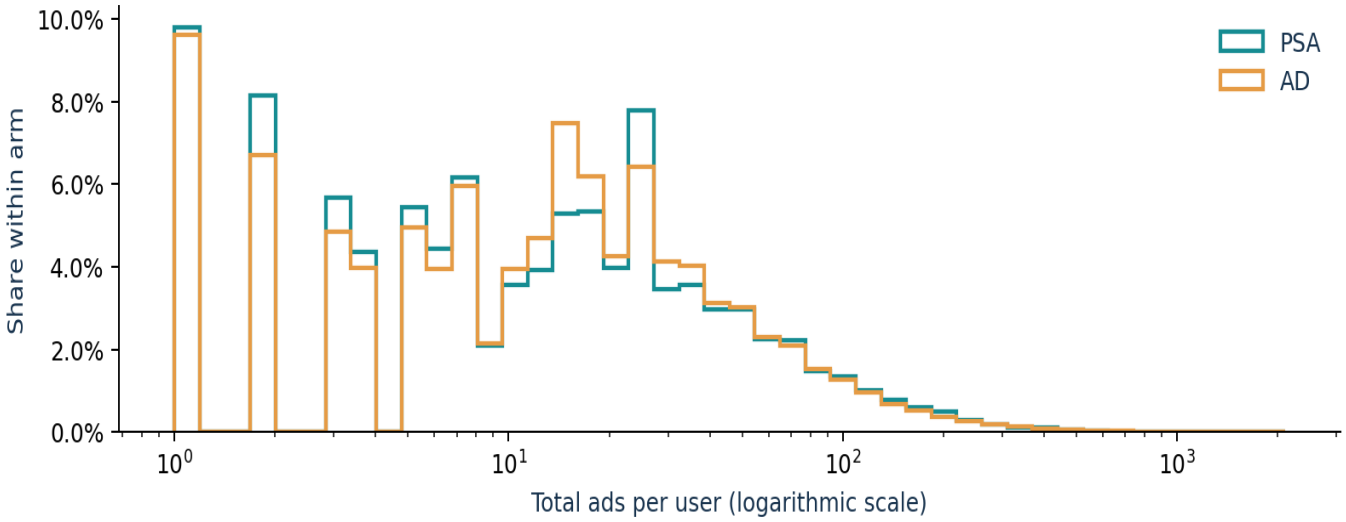
Break-even rule

Break-even CPM = incremental conversions x margin / ad impressions x 1,000. Here, that is about **0.310 x contribution margin**. A margin of 25 implies a point-estimate break-even CPM of 7.75; using the lower lift limit gives 5.92. Confirm real costs and measurement before using either boundary.

Decision implication

Conversion efficacy is promising; a spend decision requires incremental contribution to exceed incremental campaign cost under a defensible causal design. The PSA comparison measures ads versus PSA exposure, not necessarily ads versus no marketing.

05 / Exposure distribution and the long tail



Arm	Mean ads	Median	90th pct	99th pct	Maximum
PSA	24.76	12	59	206	907
AD	24.82	13	56	201	2,065

Concentration matters for spend interpretation

The common 99th percentile is 202 ads. Only 5,810 users lie strictly above it, but they account for 12.99% of all recorded exposures. Averages alone hide this long tail; impression costs can be concentrated in a small user subset.

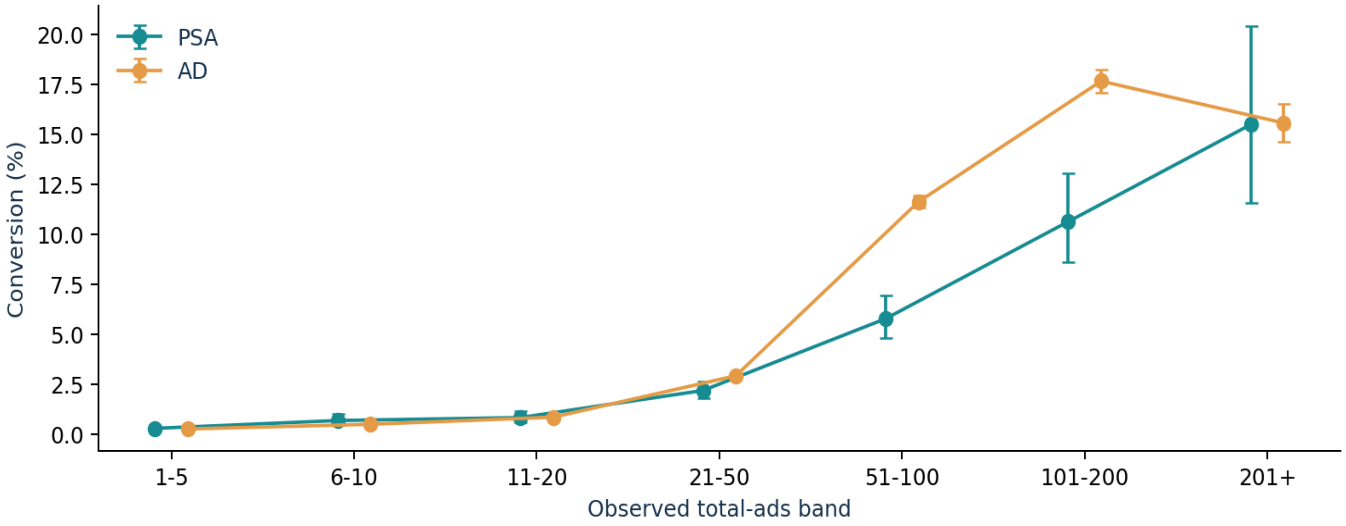
Why these are not automatic exclusions

High exposure can be genuine repeated delivery. Removing users because they received many impressions selects on observed behavior or delivery after assignment. The primary comparison therefore retains the complete cohort. Any frequency cap recommendation needs a randomized policy test, not an arbitrary percentile cutoff.

Why not adjust the main test for exposure?

A regression controlling for total ads, peak day or peak hour could block part of the treatment pathway or induce selection bias. Without timing and delivery context, those controls do not identify a better causal estimate. This project uses exposure variables for exploration rather than primary covariate adjustment.

06 / Frequency: an association, not a dose rule



Ads band	PSA users	Ad users	PSA CVR	Ad CVR	Difference
1-5	7,861	169,962	0.28%	0.25%	-0.029 pp
6-10	3,415	79,537	0.67%	0.49%	-0.188 pp
11-20	4,150	123,334	0.82%	0.84%	+0.021 pp
21-50	5,235	125,541	2.18%	2.92%	+0.738 pp
51-100	1,853	44,149	5.77%	11.63%	+5.857 pp
101-200	752	16,360	10.64%	17.68%	+7.039 pp
201+	258	5,694	15.50%	15.58%	+0.074 pp

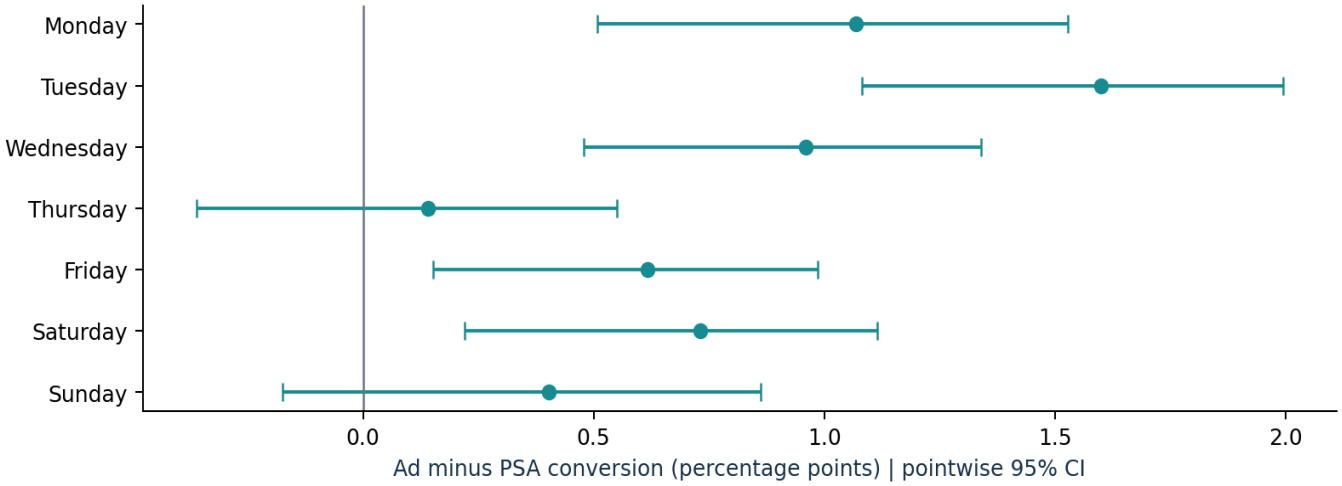
What the pattern shows

Ad-arm conversion rises from 0.25% in the 1-5 band to 17.68% in the 101-200 band, then is 15.58% at 201+. PSA conversion also rises sharply, reaching 15.50% at 201+. The apparent dose-response is therefore not exclusive to product advertising. It can reflect engagement, retargeting, repeated visits or longer time at risk. Exposure counted after purchase can reverse the apparent temporal ordering.

What would answer the operational question

Randomize eligible users to frequency-cap policies, such as different maximum deliveries, while keeping outcome follow-up fixed. Analyze assigned policies, record delivered exposure separately, and compare incremental contribution and negative-experience guardrails. The chart alone does not identify an optimal cap or prove ad fatigue.

07 / Weekday patterns and uncertainty



Peak weekday	PSA CVR	Ad CVR	Lift (pp)	PSA users	Holm p
Monday	2.26%	3.32%	+1.068	3,502	0.00956
Tuesday	1.44%	3.04%	+1.599	2,907	1.86e-06
Wednesday	1.58%	2.54%	+0.960	3,490	0.00582
Thursday	2.02%	2.16%	+0.141	3,905	1
Friday	1.63%	2.25%	+0.616	3,803	0.253
Saturday	1.40%	2.13%	+0.731	2,858	0.173
Sunday	2.06%	2.46%	+0.402	3,059	1

Read the intervals before ranking weekdays

3 of seven within-weekday ad-versus-PSA contrasts remain significant after the shared 31-test Holm correction. Tuesday has the largest observed difference (+1.599 pp); Thursday has the smallest (+0.141 pp). The intervals above are pointwise 95% intervals, not simultaneous intervals. A larger point estimate is not itself proof of a larger causal effect.

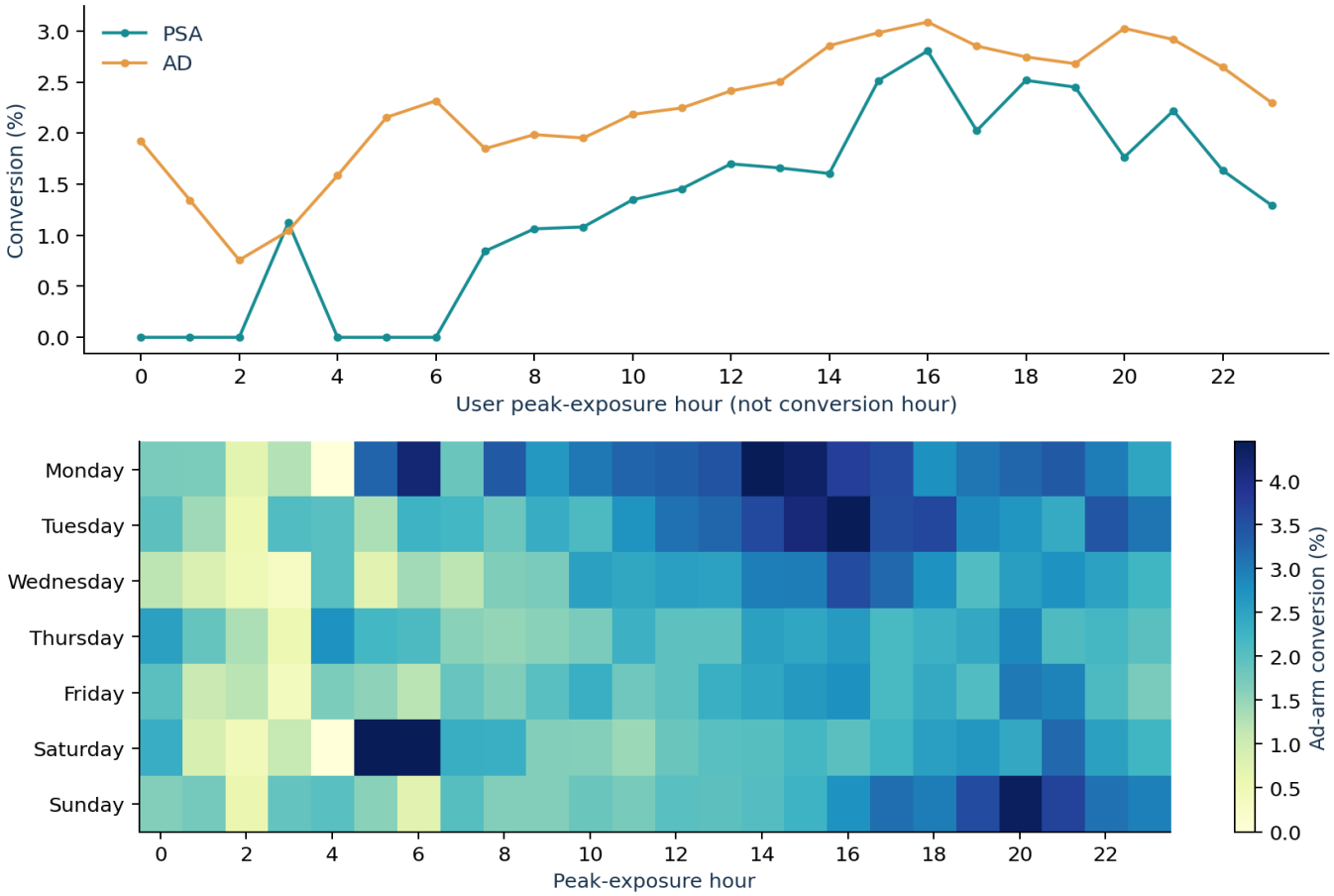
Selection and timing limits

Peak weekday is selected from the user's realized exposure history. It is neither a randomly assigned delivery day nor the purchase day. Different groups within a weekday can contain different exposure histories. Use these patterns to formulate scheduling experiments, not to shift budget causally.

Missing calendar information

There is no actual date, assignment time or conversion timestamp. We cannot estimate a weekly time trend, cumulative stopping path, seasonality, conversion lag or day-specific spend from this file.

08 / Hour patterns and day-hour exploration



Interpretation

Hourly rates vary, and 1 of 24 within-hour contrasts survive the 31-test Holm correction. Overnight control cells contain relatively few conversions; fine-grained comparisons can be noisy. The ad-arm heatmap emphasizes observed conversion context, not incremental effect or profit by slot.

No winning-hour claim

Peak-exposure hour is not the hour of conversion. Selecting the maximum among 24 hours or 168 day-hour cells creates a winner's curse. No day-hour cell testing is used to designate an optimal schedule. The companion workbook includes cell denominators and sparse-cell diagnostics so the visual can be audited.

Actionable follow-up

Test a small number of scheduling policies prospectively, randomize before delivery, and record eligibility, cost and fixed follow-up. Do not treat the heatmap as a measured return-on-ad-spend surface.

09 / Multiple testing and heterogeneity

A separate primary claim and an exploratory family

The overall arm comparison is the primary test. The seven weekday and 24 hour comparisons form one exploratory family of 31 two-sided Fisher exact tests. Exact tests avoid relying on a z approximation in sparse hourly cells. Holm's step-down adjustment is applied across the entire family and controls family-wise error under valid individual p-values, including dependence among contrasts.

Within-segment significance does not establish effect differences

A significant ad-versus-PSA contrast on Monday and a non-significant contrast on Tuesday do not prove a Monday-versus-Tuesday difference. The correct diagnostic asks whether treatment effects vary across categories. We compare a common-treatment binomial logistic model with a model allowing category-by-treatment interaction.

Exploratory interaction	LR statistic	df	Raw p	Holm p (2 tests)
peak_day	16.220	6	0.0126	0.0252
peak_hour	29.100	23	0.177	0.177

Interaction scale and interpretation

The weekday interaction remains detectable after correcting two interaction tests (adjusted p = 0.025), whereas the hourly interaction does not (p = 0.177). These tests assess heterogeneity on the log-odds scale, not directly on the absolute-lift scale. The unrestricted grouped model is saturated. Rare cells make asymptotic results less dependable, particularly for hours.

Design limits still dominate

Even an interaction signal cannot turn realized exposure categories into randomized subgroups. It is exploratory evidence of different observed patterns and may reflect selection, delivery or user behavior. Follow-up experiments are needed to identify causal scheduling effects.

No significance fishing

No frequency band was selected to maximize lift. No 168-cell heatmap search is used for confirmatory claims. No predictive classifier is substituted for treatment-effect estimation, and no post-treatment adjustment is presented as a causal correction.

10 / Robustness and influence diagnostics

Method / cohort	Estimate or p-value	What it checks
Pooled two-proportion z	1.71e-13	Primary large-sample null test
Pearson chi-square, no correction	1.71e-13	Equivalent 2x2 null formulation
Fisher exact, two-sided	1.05e-14	Conditional exact-test cross-check
Newcombe 95% CI	[0.587, 0.936] pp	Primary uncertainty method
Wald 95% CI	[0.595, 0.943] pp	Simple interval sensitivity
Binomial bootstrap 95% CI	[0.593, 0.941] pp	100,000 independent arm draws
Exposure-trimmed lift	+0.792 pp	Users with total ads <= p99

What agrees

The primary tests and alternative interval calculations support the same positive direction. The binomial resampling uses observed arm probabilities and fixed arm sizes, with seed 20261101. For independent Bernoulli observations, this reproduces the distribution obtained by resampling outcomes within each arm; it does not model assignment failures, clustering or outcome mismeasurement.

What trimming can and cannot tell us

Excluding users above 202 exposures leaves 582,291 users and a difference of +0.792 pp, with a 95% interval [0.616, 0.952] pp. Directional persistence suggests the overall association is not solely carried by the extreme tail.

However, this is a changed population defined after exposure. Its effect size need not equal the original estimand. It cannot establish what would happen if a frequency cap were imposed. The complete cohort remains primary.

Unmeasured threats

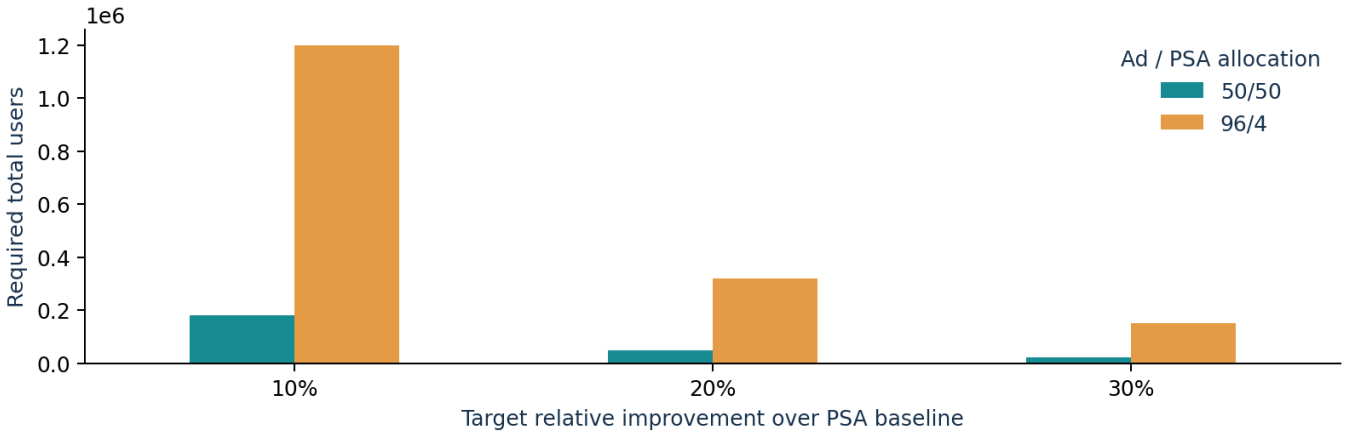
Neither a second p-value nor a bootstrap can repair unverified randomization, spillovers between users, different attribution windows, missing eligible non-exposed users or outcome tracking differences. Those require experimental logs or a new controlled study.

11 / Prospective power and a better next trial

Planning assumptions

Use the observed PSA rate (1.785%) as a planning baseline, 80% power and two-sided alpha 0.05. Target improvements of 10%, 20% and 30% relative to that baseline. These targets are illustrative business choices, not a retrospective claim about what this experiment was powered to detect.

Relative target	Absolute target	Allocation	PSA needed	Ad needed	Total
10%	0.179 pp	50/50	90,587	90,587	181,174
10%	0.179 pp	96/4	47,971	1,151,304	1,199,275
20%	0.357 pp	50/50	23,703	23,703	47,406
20%	0.357 pp	96/4	12,731	305,544	318,275
30%	0.536 pp	50/50	11,003	11,003	22,006
30%	0.536 pp	96/4	5,982	143,568	149,550



Trade-off

An approximately balanced design generally needs fewer total users for comparable precision when outcome variability and user cost are similar. A small holdout can reduce exposure opportunity cost, but requires more total traffic. Allocation should be a planned trade-off rather than inferred from the final file.

Trial specification

Define eligibility and a meaningful incremental-conversion or contribution threshold. Randomize stable users before delivery; preserve assignment even when exposure fails. Specify a fixed outcome window, follow users equally, log exposure and conversion times, and set a minimum duration covering complete weekly cycles. Do not stop on the first nominally significant result; use a fixed horizon or a prespecified sequential method.

Guardrails and follow-up

Record spend, contribution margin, user complaints, opt-outs, reach and delivery failures. Test frequency caps or scheduling policies in separate randomized comparisons with a small prespecified set of contrasts and appropriate multiplicity control.

12 / Decision, limitations and reproducibility

Decision area	Evidence-supported conclusion	Next action
Conversion efficacy	Positive full-cohort ad-versus-PSA difference	Validate assignment and measurement
Campaign scale	Economic return is not observed	Obtain CPM and contribution margin
Frequency optimization	Exposure-associated conversion patterns	Randomize cap policies
Scheduling optimization	Peak-context patterns and uncertainty	Randomize schedule policies
Data quality	No missing or duplicate users in this file	Audit eligibility and logging upstream

Material limitations

No experiment dates, assignment logs, prespecified allocation, conversion window, costs, revenue, baseline covariates, demographic segments or operational guardrails are supplied. No historical stability or long-term effect can be established. User-level independence, valid assignment and comparable measurement remain assumptions.

Reproduce the work

Run marketing/src/analyze.py, then marketing/src/report.py and marketing/src/dashboard_v4.py from the repository root. The raw download is preserved. Cleaned records, SQLite tables, executed SQL, all query outputs, segment statistics, economic scenarios, sample-size plans, seeds and dashboard aggregates accompany the report. The dashboard uses an embedded aggregate snapshot and does not refresh directly from Kaggle.

Source and method references

Kaggle / Favio Vazquez: Marketing A/B Testing

NIST: Wilson and related proportion confidence intervals

NIST: Difference of proportions hypothesis test

Statsmodels source: independent-proportion intervals and power formulas

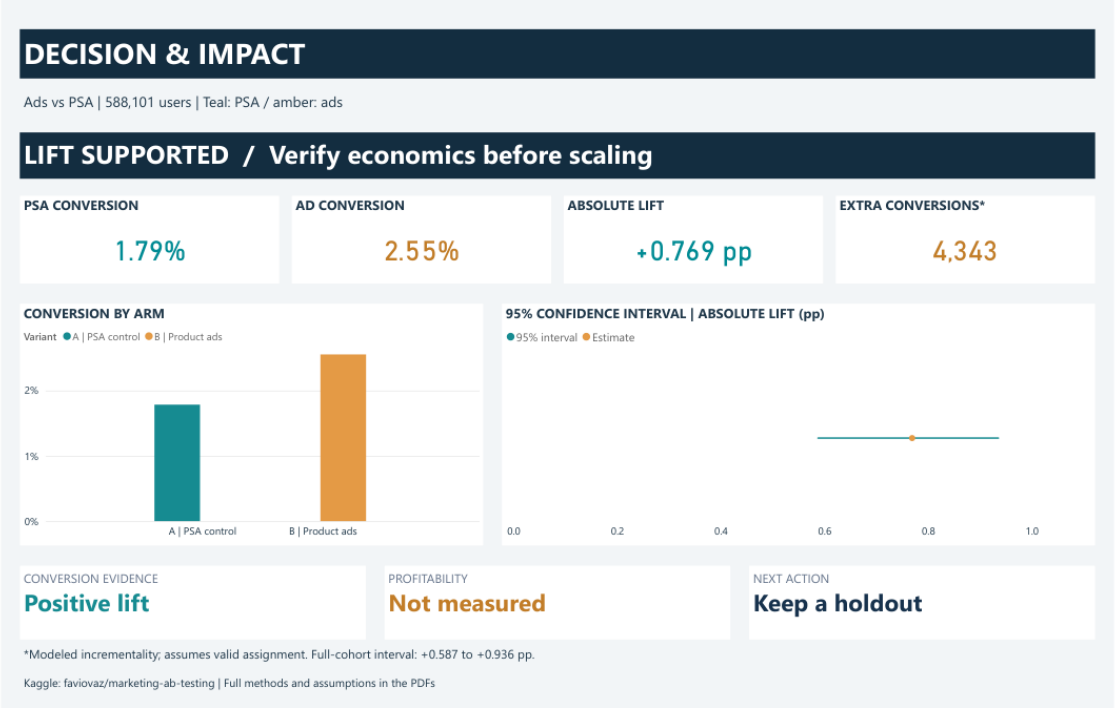
Newcombe RG (1998). Interval estimation for the difference between independent proportions: comparison of eleven methods. Statistics in Medicine 17:873-890. Holm S (1979). A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6:65-70.

Raw CSV SHA-256: de6ce8def1a6559e9e5ab364fce07cfe2e84f02dd407c3586eea55743c7e17f2

Analysis prepared 01 November 2026; final dashboard alignment updated 02 November 2026. The final five-page Power BI export is included as Marketing_AB_Dashboard_Final.pdf.

13 / Final dashboard and decision view

Final export: 02 November 2026



Read the decision before the descriptive charts

The final opening page prioritizes the full-cohort comparison: PSA conversion 1.79%, ad conversion 2.55%, absolute lift +0.769 percentage points and 4,343 modeled additional conversions. The interval chart spans +0.587 to +0.936 percentage points. Rounded cards reconcile with the higher-precision estimates in this report.

Five pages, distinct analytical purposes

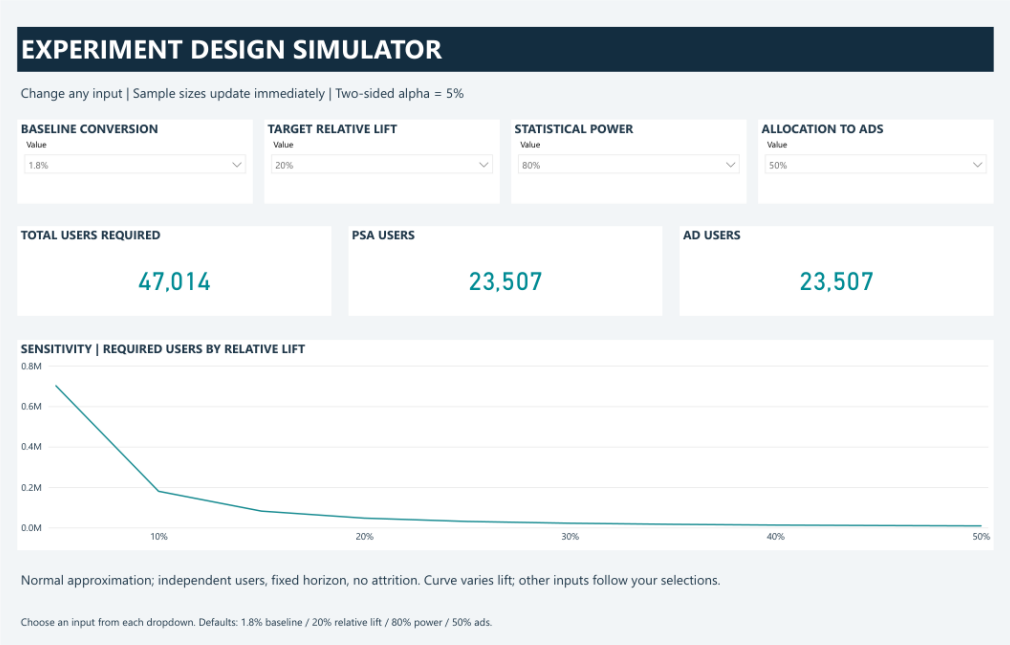
Page	Purpose
1 / Decision & impact	Conversion evidence, modeled incrementality and business decision
2 / Experiment design simulator	Four selectable planning inputs; live sample-size recalculation
3 / Evidence	Primary p-value, risk ratio, allocation and holdout variance
4 / Exposure explorer	Observed frequency and concentration; explicitly non-causal
5 / Timing explorer	Descriptive peak-day/hour patterns and denominators

Interpretation guardrail

Page 4 prominently states that more exposure does not prove more conversions. Engagement and targeting can affect both variables. Frequency and timing charts generate hypotheses for randomized policy trials; they do not authorize a causal dose or scheduling recommendation.

The PDF is a fixed snapshot. Use the PBIX file for interactive controls and filtering.

14 / Interactive experiment-design simulator



Inputs and default results

Baseline conversion, target relative lift, statistical power and allocation to ads are native single-selection dropdowns. Defaults are 1.8%, 20%, 80% and 50%, respectively. The two-sided significance level stays fixed at 5%. Required total, PSA and ad sample sizes recalculate immediately; the sensitivity curve varies lift from 5% to 50% while using the selected baseline, power and allocation.

Scenario (other inputs at defaults)	PSA users	Ad users	Total
Default	23,507	23,507	47,014
Power = 90%	31,469	31,469	62,938
Relative lift = 40%	6,399	6,399	12,798
Ad allocation = 96%	12,626	303,024	315,650
Baseline = 2.0%	21,109	21,109	42,218

Why these differ from the earlier planning table

The report scenarios use the observed PSA rate of 1.7854%; the dashboard default uses the selectable rounded baseline of 1.8%. Both use the same normal-approximation formula. The dropdowns contain discrete values, not unrestricted numeric entry. This is prospective planning, not retrospective observed power.

Assumptions: independent users, fixed horizon, equal follow-up, no attrition, clustering or repeated testing. Duration requires eligible traffic and follow-up inputs.